

Dynamics – Project 5: Pendulum

Objective

For this project, we were in a group of 3 and the objective was to measure the frequency, f , and period, T , (time) of our pendulum motions given a known length of lanyard. We were to find the relationship between lengths and frequencies using known and varying lanyard lengths.

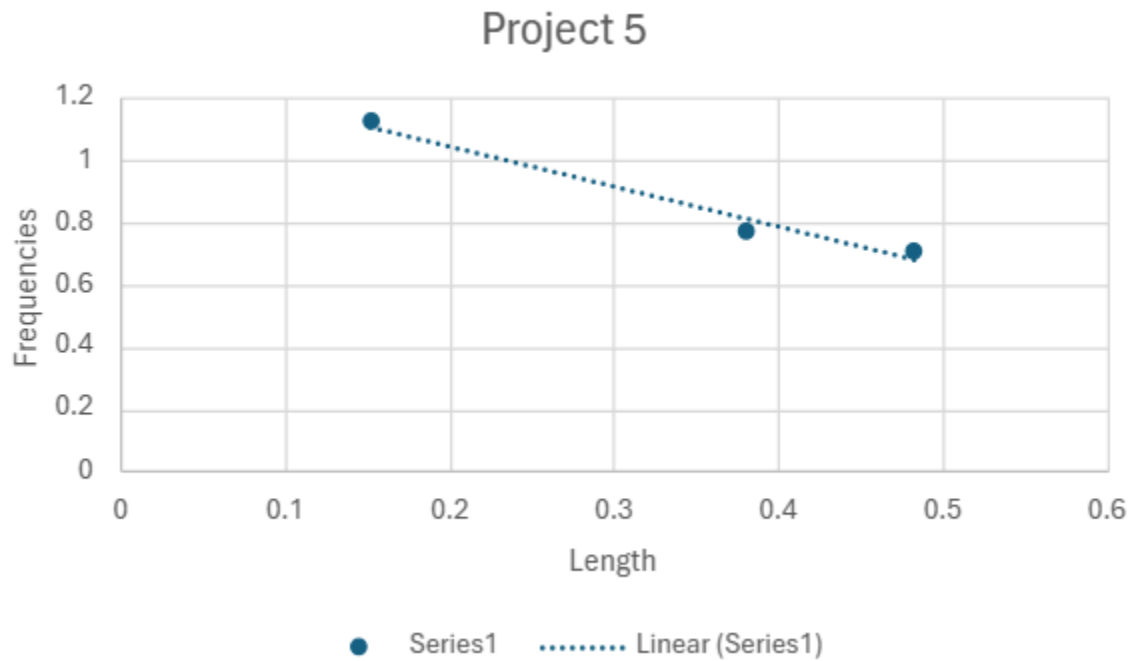
Methodology

We used car keys on a lanyard as our pendulum. We measured the full-length L of the lanyard then shortened it by holding lower on the lanyard and measuring again. We also stopwatch to measure the time it takes for one cycle to get the period as a comparison to our calculations. We knew it was one cycle by releasing the keys from Cole's hand then waiting for them to return. We also calculated the period and frequency using the given equations:

$$T = 2\pi \cdot \sqrt{\frac{L}{g}}$$
$$f = \frac{1}{T}$$

Analysis

Length	Measured Time	Expected Time	Percent Difference	Measured Frequency	Calculated Frequency
0.4826	1.41	1.393602842	1.162918981	0.709219858	0.717564553
0.381	1.3	1.23824862	4.750106187	0.769230769	0.807592259
0.1524	0.89	0.783137189	12.00705736	1.123595506	1.27691548
Average	Average	Average	Average	Average	Average
0.338666667	1.2	1.13832955	5.973360843	0.867348711	0.934024097



Hand computational work:

Handwritten calculations on lined paper showing the determination of period (T) and frequency (f) for three different pendulum lengths. The calculations are as follows:

$$T_1 = 2\pi \sqrt{\frac{0.4826 \text{ m}}{9.81 \frac{\text{m}}{\text{s}^2}}} = 2\pi \sqrt{0.0492 \text{ s}^2} = 1.39 \text{ s}$$
$$T_2 = 2\pi \sqrt{\frac{0.381 \text{ m}}{9.81 \frac{\text{m}}{\text{s}^2}}} = 2\pi \sqrt{0.0388 \text{ s}^2} = 1.24 \text{ s}$$
$$T_3 = 2\pi \sqrt{\frac{0.1524 \text{ m}}{9.81 \frac{\text{m}}{\text{s}^2}}} = 2\pi \sqrt{0.0155 \text{ s}^2} = 0.783 \text{ s}$$

$$f_1 = \frac{1}{T_1} = 0.719 \text{ Hz}$$
$$f_2 = \frac{1}{T_2} = 0.806 \text{ Hz}$$
$$f_3 = \frac{1}{T_3} = 1.28 \text{ Hz}$$

Video:

<https://youtube.com/shorts/o5koJfDfOSE?si=UT5aW06AJFUnxEcO>

Conclusion:

Finally, we measured these three values: (**T1 = 1.41s** and **F1 = 0.709 Hz** / **T2 = 1.30s** and **F2 = 0.769 Hz** / **T3 = 0.89s** and **F3 = 1.123 Hz**). And, by the calculation, we obtained these three values: (**T1' = 1.39s** and **F1' = 0.719 Hz** / **T2' = 1.24s** and **F2' = 0.806 Hz** / **T3' = 0.783s** and **F3' = 1.28 Hz**).

The relation between the length of the string and the time of the swinging motion is inversely proportional. And our results are pretty good. To increase the result we measured, we can use a system to improve the accuracy of the measurement of time during the swinging motion. We can also try to make sure that the starting point is always the same for each measurement.

Picture:

